

Noise-induced shallow circuits and absence of barren plateaus

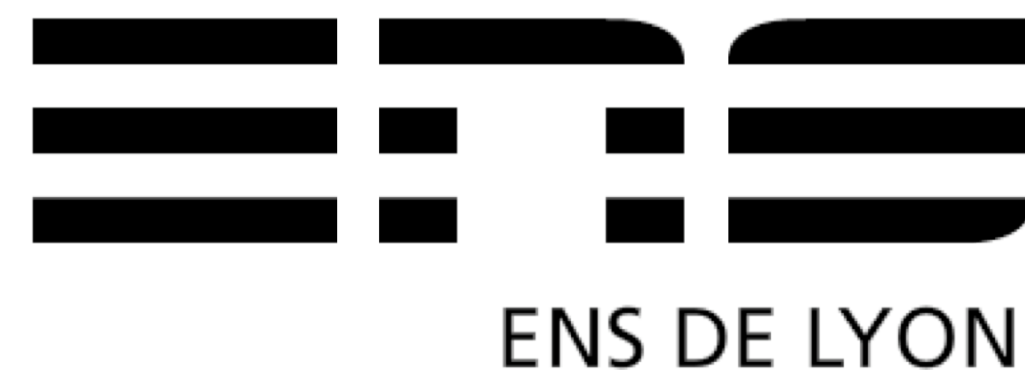
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Joint work with:

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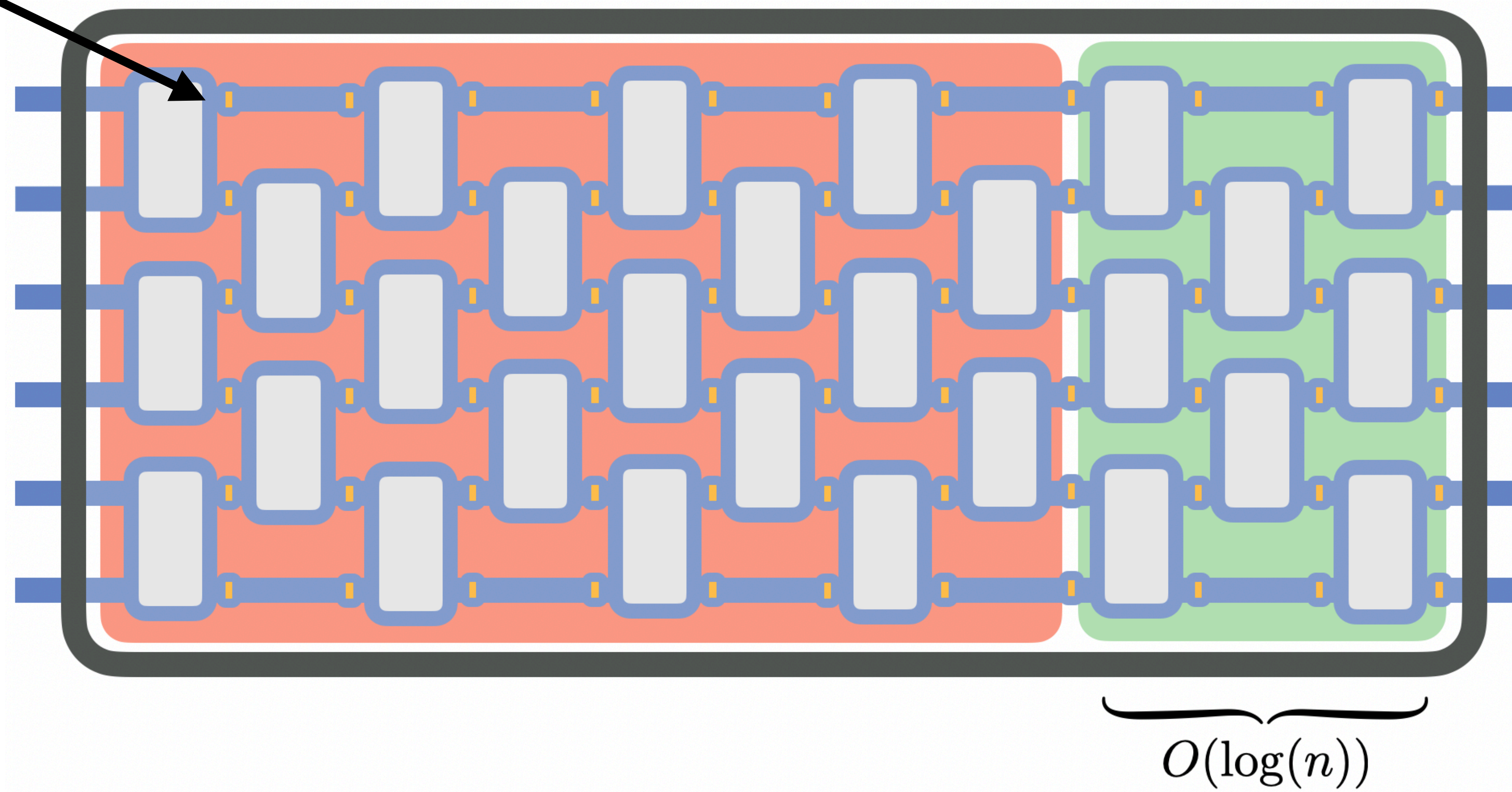


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Spoiler

(What we'll see)

'Realistic' noise



Only the last layers matter to estimate expectation values, for most circuits.

- This implies:
- Efficient classical simulation
 - Lack of Barren plateaus

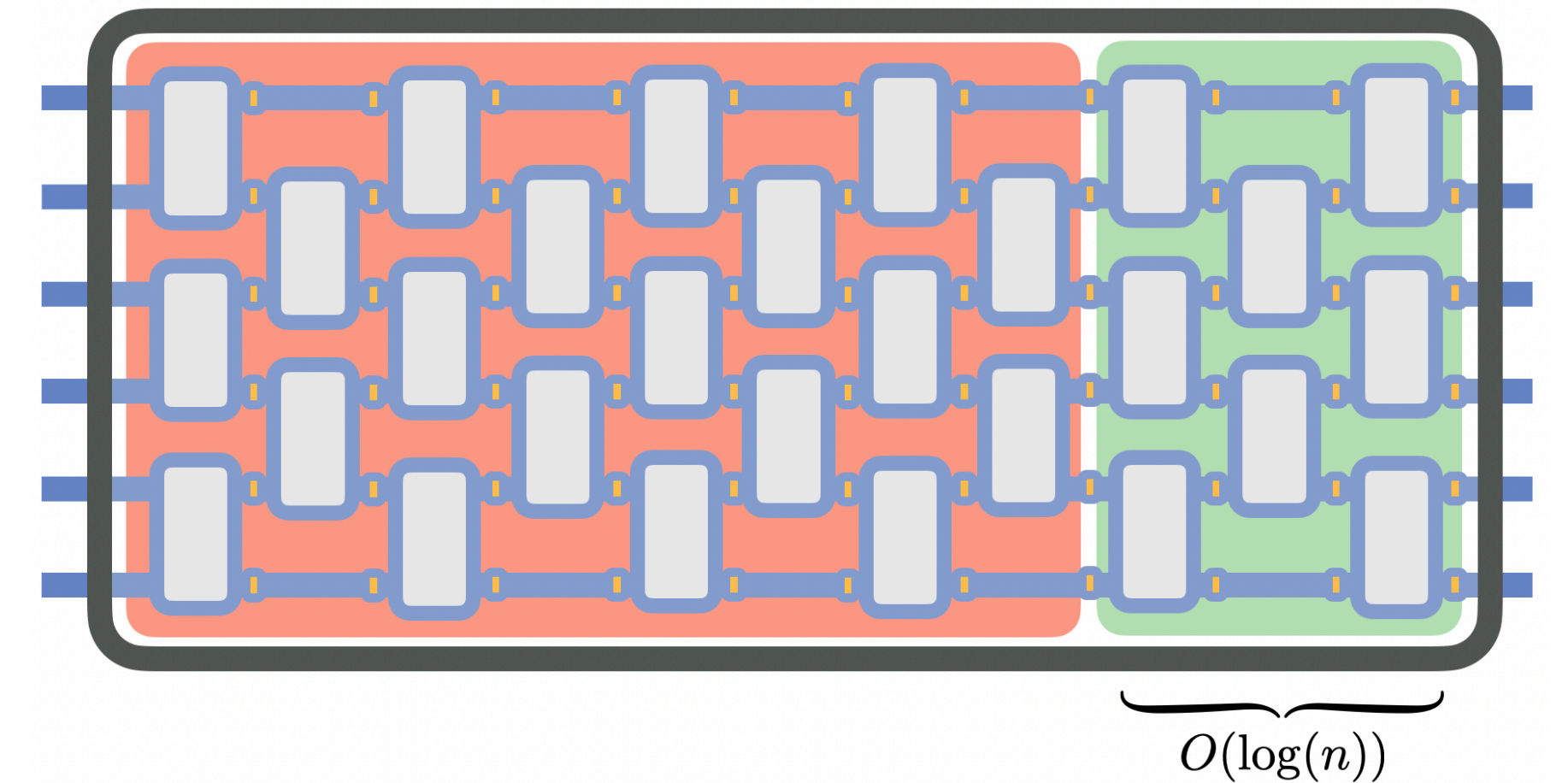
Outline

- ☼ Noise in quantum circuits

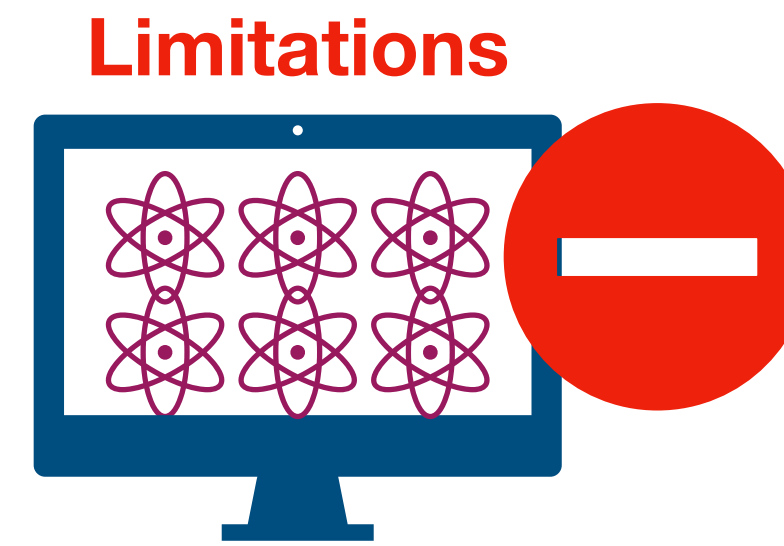
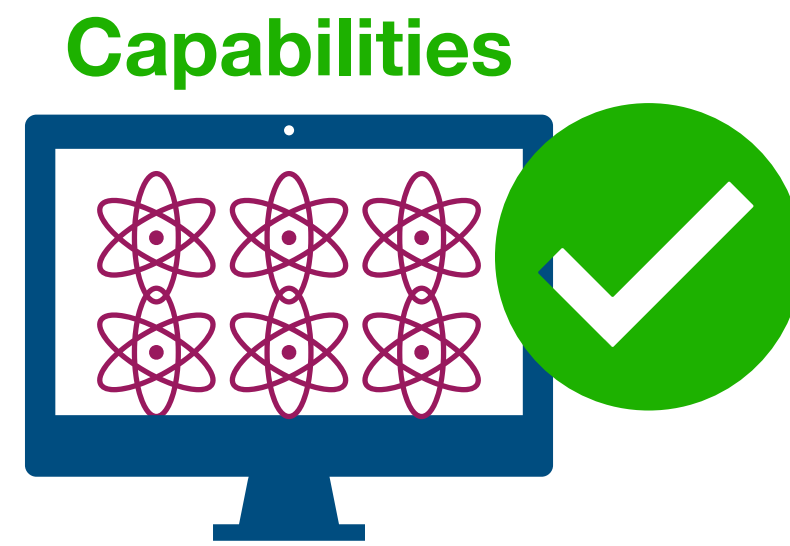
- ☼ Effective shallow circuits

- ☼ Classical simulation of Pauli expectation values of noisy random circuits

- ☼ Barren plateaus



- Understanding **noise impact** in current quantum devices is important for:



- Many previous works modeled noise as solely **depolarizing**.

$$\mathcal{N}(\sigma) = (1 - p)\sigma + p\frac{I}{2} \quad \text{with } p \in [0,1]$$

- **Depolarizing noise**, e.g., induces:
 - ✱ Barren plateaus in variational quantum algorithms [1],
 - ✱ Efficient classical simulation of ‘**supremacy**’ **sampling experiments** [2].
- But, **tiny departure** from this model could lead to **different conclusions** [3,4].

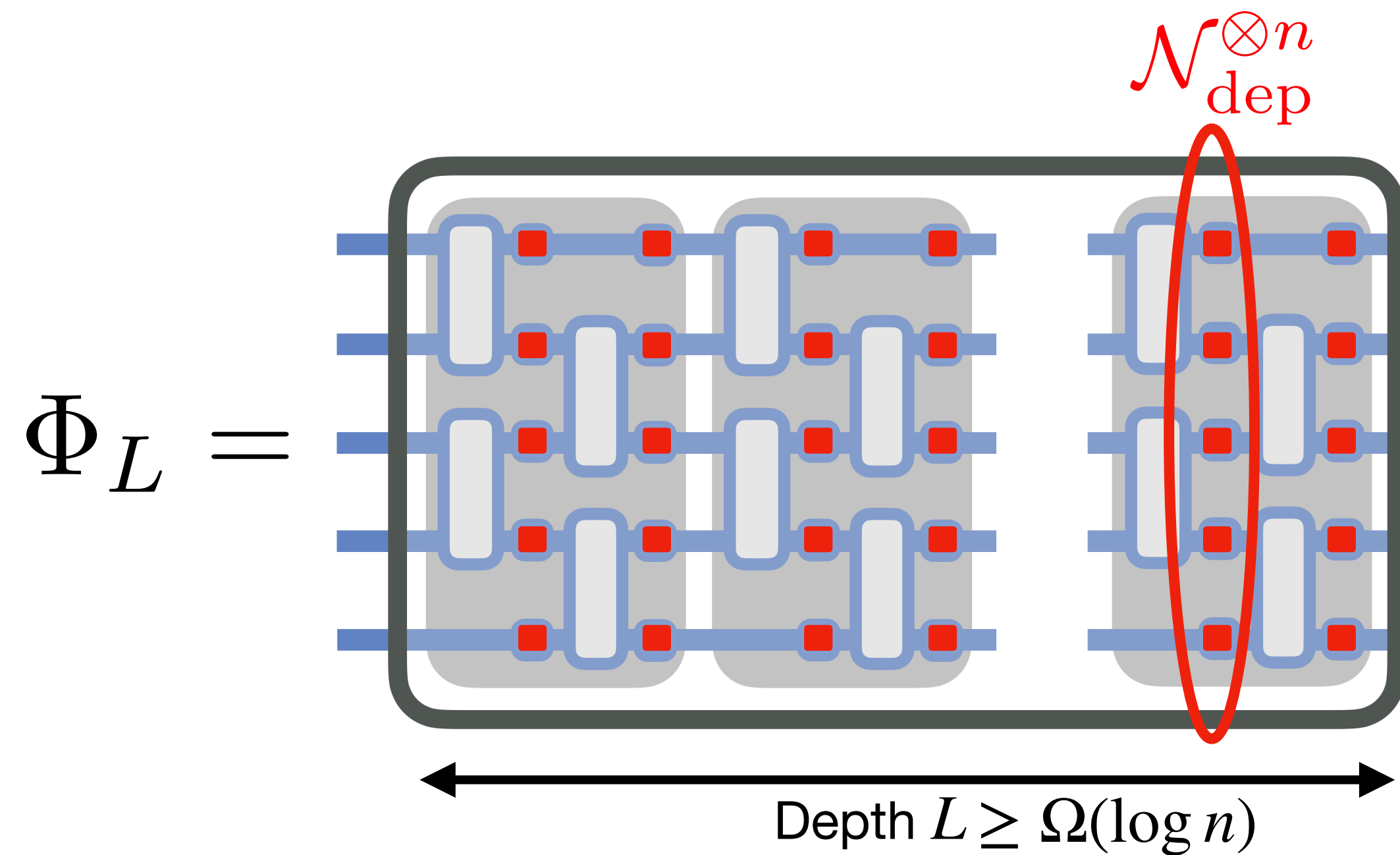
[1] Noise-induced barren plateaus in variational quantum algorithms. Wang *et al*, Nature Comm. (2021).

[2] A polynomial-time classical algorithm for noisy random circuit sampling, Aharonov *et al.*, STOC (2023)

[3] Quantum refrigerator, Ben-Or *et al*, QIP (2013).

[4] Effect of non-unital noise on random circuit sampling, Fefferman *et al*, QIP (2023).

- The **depolarizing** assumption, if true, **enormously constrains noisy computation**:



$$\left\| \Phi_L(\rho_0) - \frac{\mathbb{I}}{2^n} \right\|_1 \leq \exp(-\Theta(L)), [5]$$

□
Quantum computation possible only for $\log(n)$ depth! [6]

$$|\text{Tr}(P\Phi_L(\rho_0))| \leq \exp(-\Theta(L)),$$

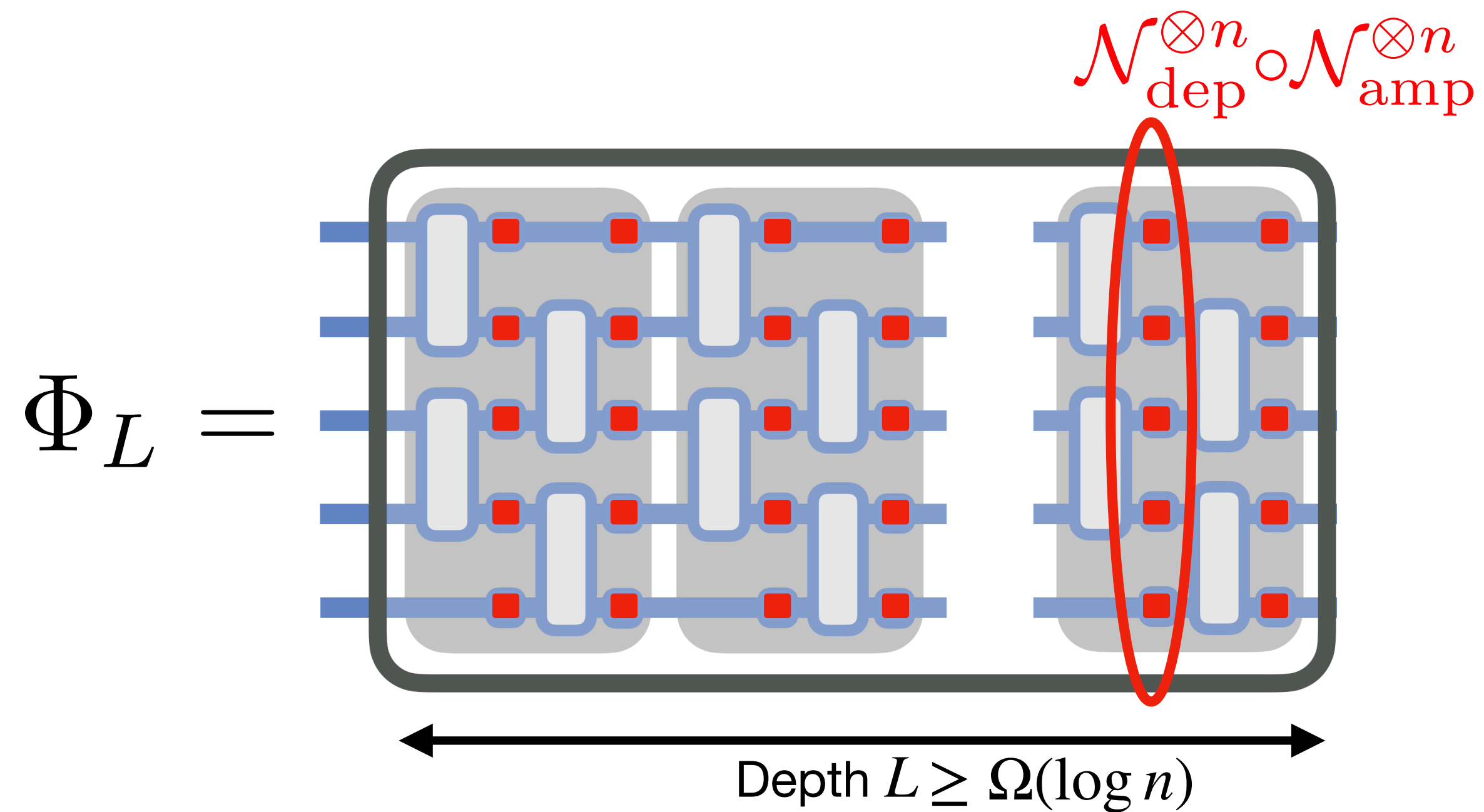
for any Pauli P and input state ρ_0

- Thus, **efficient** classical simulation of **all** deep depolarizing-noisy circuits is “trivial”.

[5] Relative entropy convergence for depolarizing channels, Muller-Hermes *et al*, J. Math. Phys. (2016).

[6] Limitations of noisy reversible computation, Aharonov *et al*, ArXiv (1996).

- The **depolarizing** assumption, if true, **enormously constrains noisy computation**:



$$\left\| \Phi_L(\rho_0) - \frac{\mathbb{I}}{2^n} \right\|_1 \leq \exp(-\Theta(L)), \quad [5]$$

Quantum computation possible only for $\log(n)$ depth! [6]

However, for noises different from depolarizing, not all circuits become trivial after log-depth!

- Real **hardware noise** is **not solely depolarizing** (which is ‘unital’, i.e., $\mathcal{N}(I) = I$)
- Other noise components present (T1-error, amplitude damping, ...) are **non-unital**.

[3] Quantum refrigerator, Ben-Or *et al*, QIP (2013). $\longrightarrow$

‘Non-unital’ noisy circuits can be made fault-tolerant, without fresh auxiliary qubits!

[5] Relative entropy convergence for depolarizing channels, Muller-Hermes *et al*, J. Math. Phys. (2016).

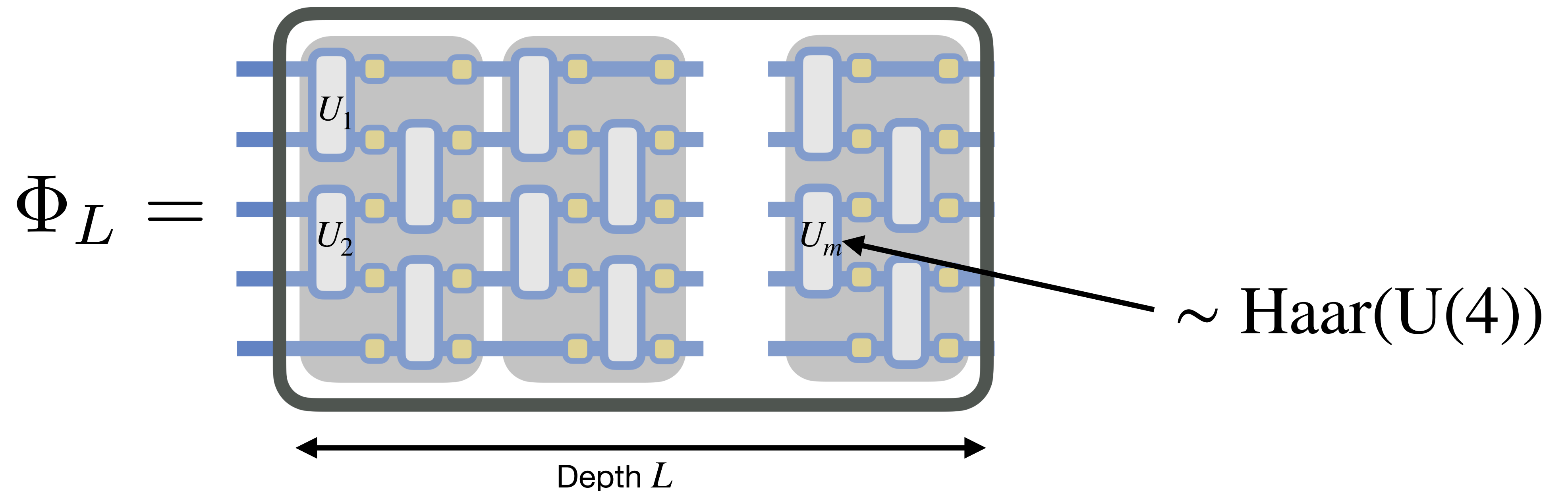
[6] Limitations of noisy reversible computation, Aharonov *et al*, ArXiv (1996).

Thus, if the noise is non-unital:
one can carefully engineer the circuit to perform reliable quantum computation [3]

(If $BQP \neq BPP$, Pauli expectation values of **all** non-unital noisy circuits cannot be efficiently estimated classically.)

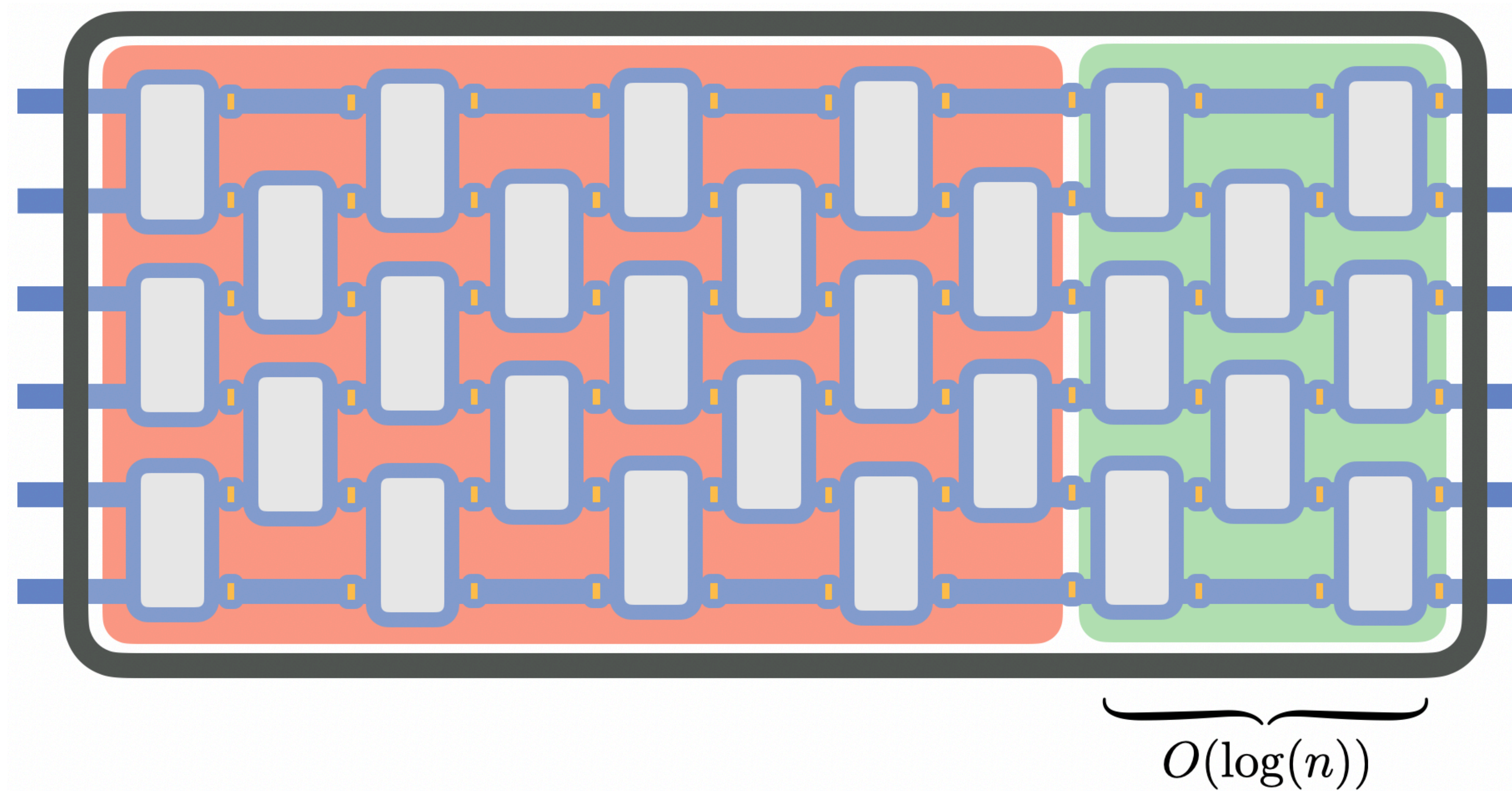
But, what happens for generic noisy circuits?

(Our work)



[3] Quantum refrigerator, Ben-Or *et al*, QIP (2013).

What happens for generic noisy circuits?

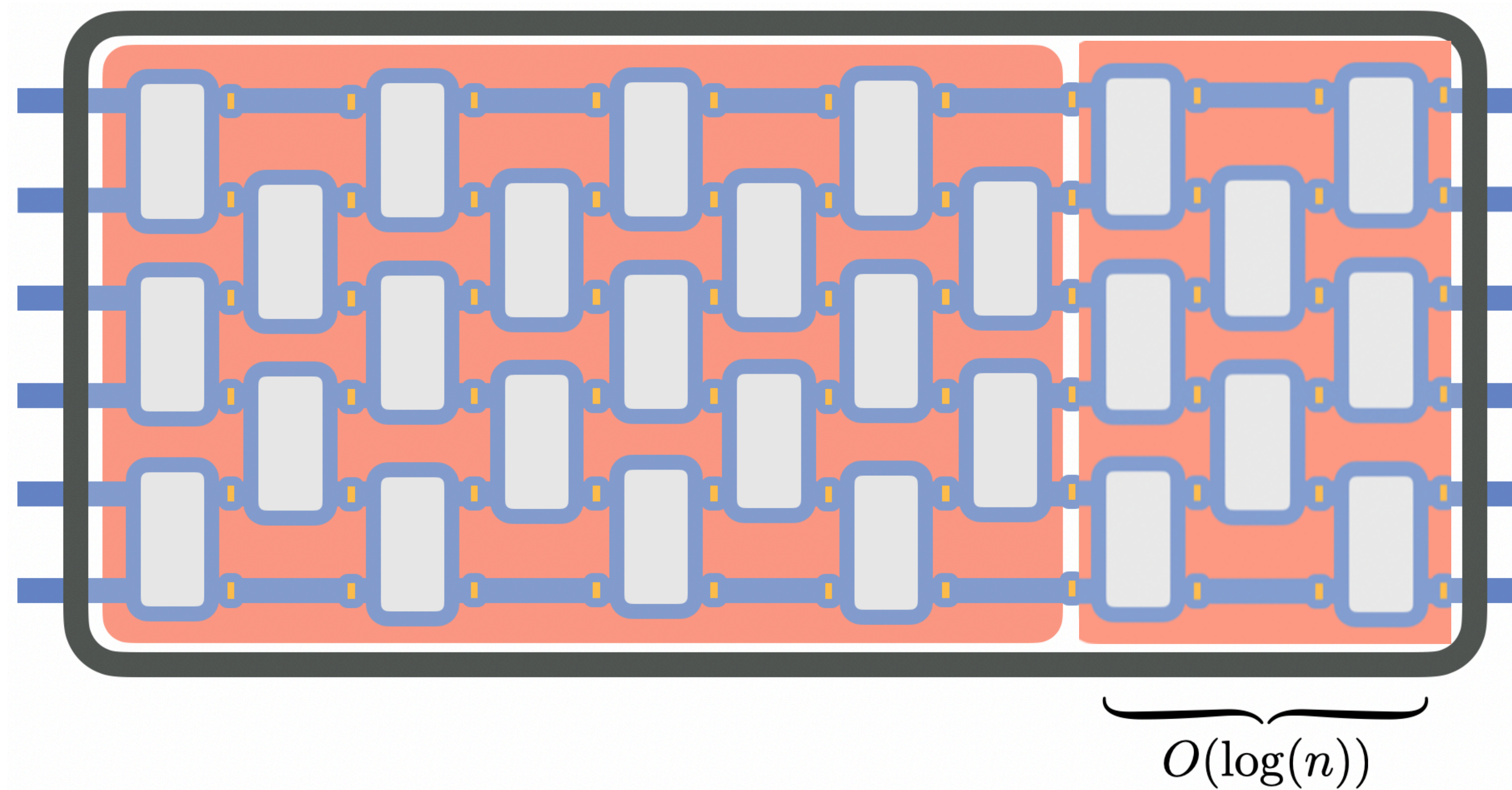


Only the last $\log(n)$ layers matter to estimate expectation values, for most circuits.

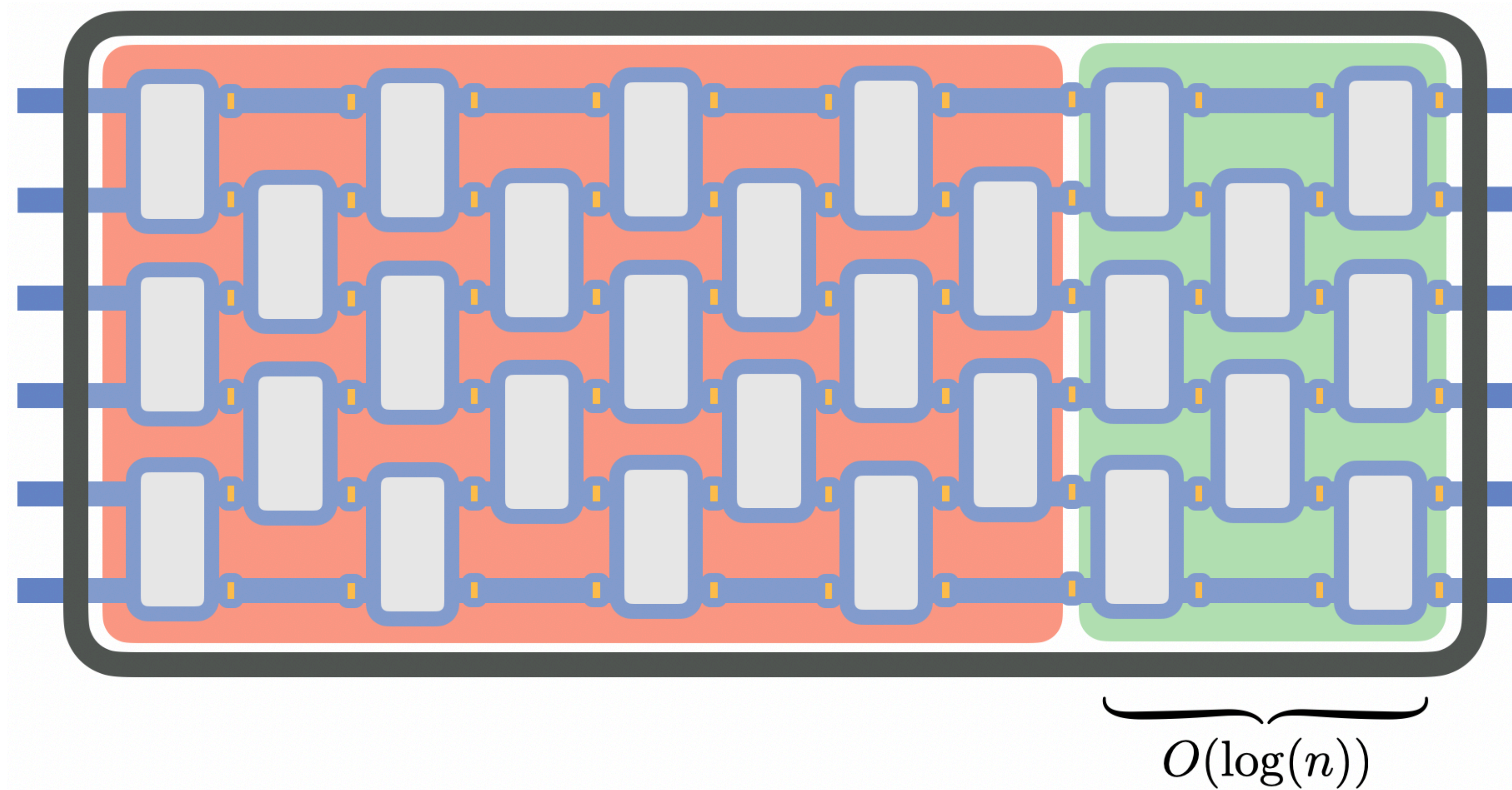
Intuition: the content of the first layers is forgotten, if nothing *clever* is done to remember!

If the noise is depolarizing, no layers matter.

(Even for ALL circuits)



If the noise is **non-unital**, the last $O(\log(n))$ do matter!
(for most circuits)



Effective shallow circuits

Proposition.

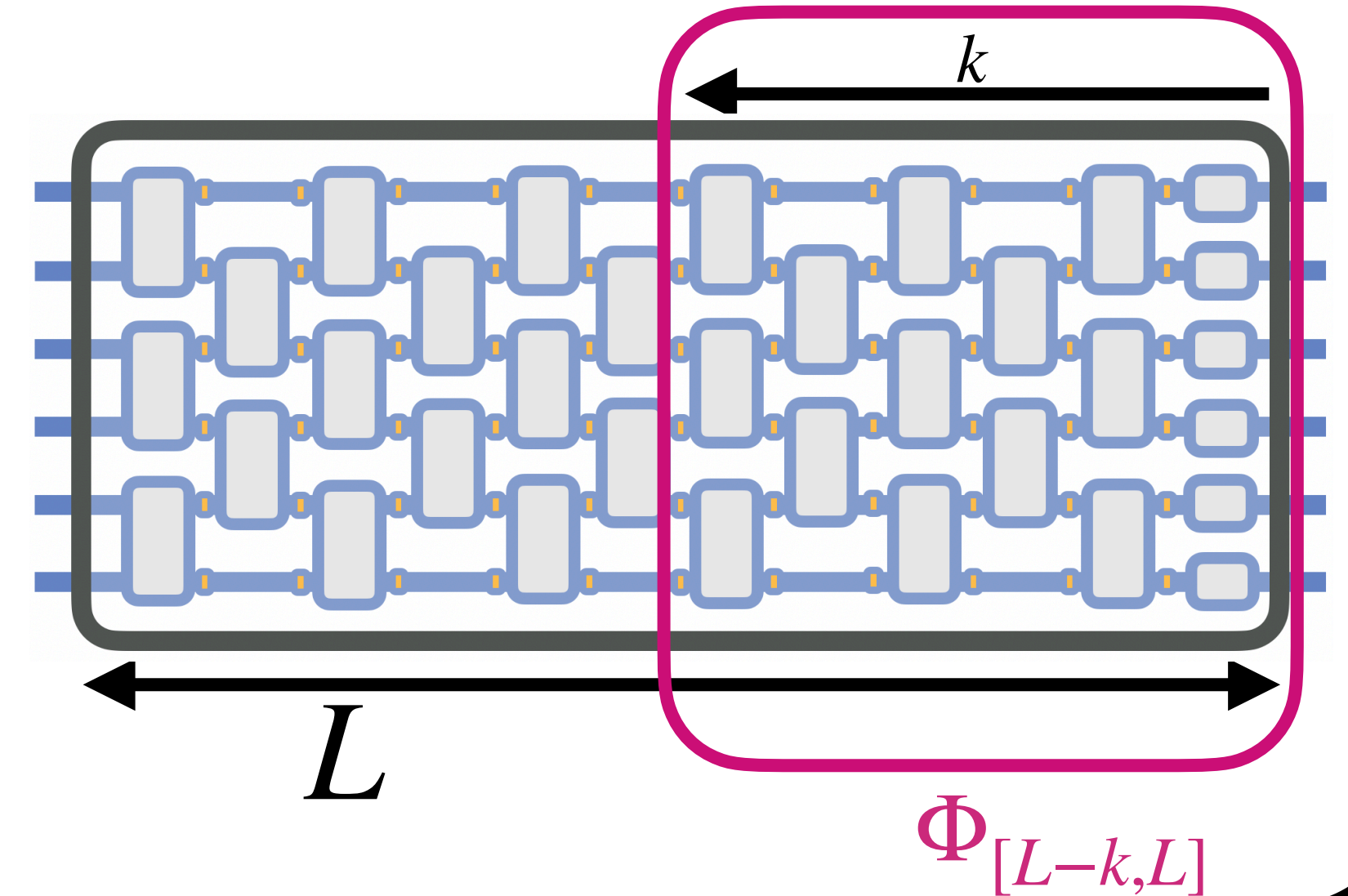
For any initial state ρ_0 (possibly complex) and observable O ,

$$\mathbb{E}_{\Phi} \left[\left| \text{Tr} (O\Phi(\rho_0)) - \text{Tr} (O\Phi_{[L-k,L]}(\sigma_0)) \right| \right] \leq \|O\|_{\infty} \exp(-\Omega(k)). \quad \Phi =$$

$\nwarrow$
 Target expectation value

$\nwarrow$
 Last k layers

$\nwarrow$
 Any preferred initial state



This follows from the following more general theorem:

Theorem.

For any k depth of the circuit Φ_k , states ρ, σ , and Pauli P , we have:

$$\mathbb{E}_{\Phi_k} \left[\left| \text{Tr}(P\Phi_k(\rho)) - \text{Tr}(P\Phi_k(\sigma)) \right| \right] \leq \exp(-\Omega(k + |P|)).$$

Trace distance convergence

Theorem. (Trace distance bound)

For any $L \geq \Omega(n)$ depth of the noisy n -qubits circuit Φ , states ρ, σ , we have:

$$\mathbb{E}_{\Phi} \|\Phi(\rho) - \Phi(\sigma)\|_1 \leq \exp(-\Omega(L)).$$

- A general result like this, but without $\mathbb{E}_{\Phi}$, cannot be obtained (e.g.) because of Ref. [3] (which shows how to leverage non-unital noise to perform fault-tolerant quantum computation without fresh-auxiliary qubits.)
- However, in certain ‘high noise’ regime, we can obtain it:

Theorem. (High-noise worst-case, informal)

For any $L \geq \Omega(\log(n))$ depth of the (very) noisy n -qubits circuit Φ , states ρ, σ , we have:

$$\|\Phi(\rho) - \Phi(\sigma)\|_1 \leq \exp(-\Omega(L)).$$

Classical simulation of noisy random circuits

Task: Estimate $\text{Tr}(P\Phi(\rho_0))$, with high probability over the choice of Φ .

Solution: It suffices to output $\text{Tr}(P\Phi_{[L-k,L]}(|0^n\rangle\langle 0^n|))$.

Previous proposition.

$$\mathbb{E}_{\Phi} [|\text{Tr}(P\Phi(\rho_0)) - \text{Tr}(P\Phi_{[L-k,L]}(|0^n\rangle\langle 0^n|))|] \leq \exp(-\Omega(k + |P|)).$$

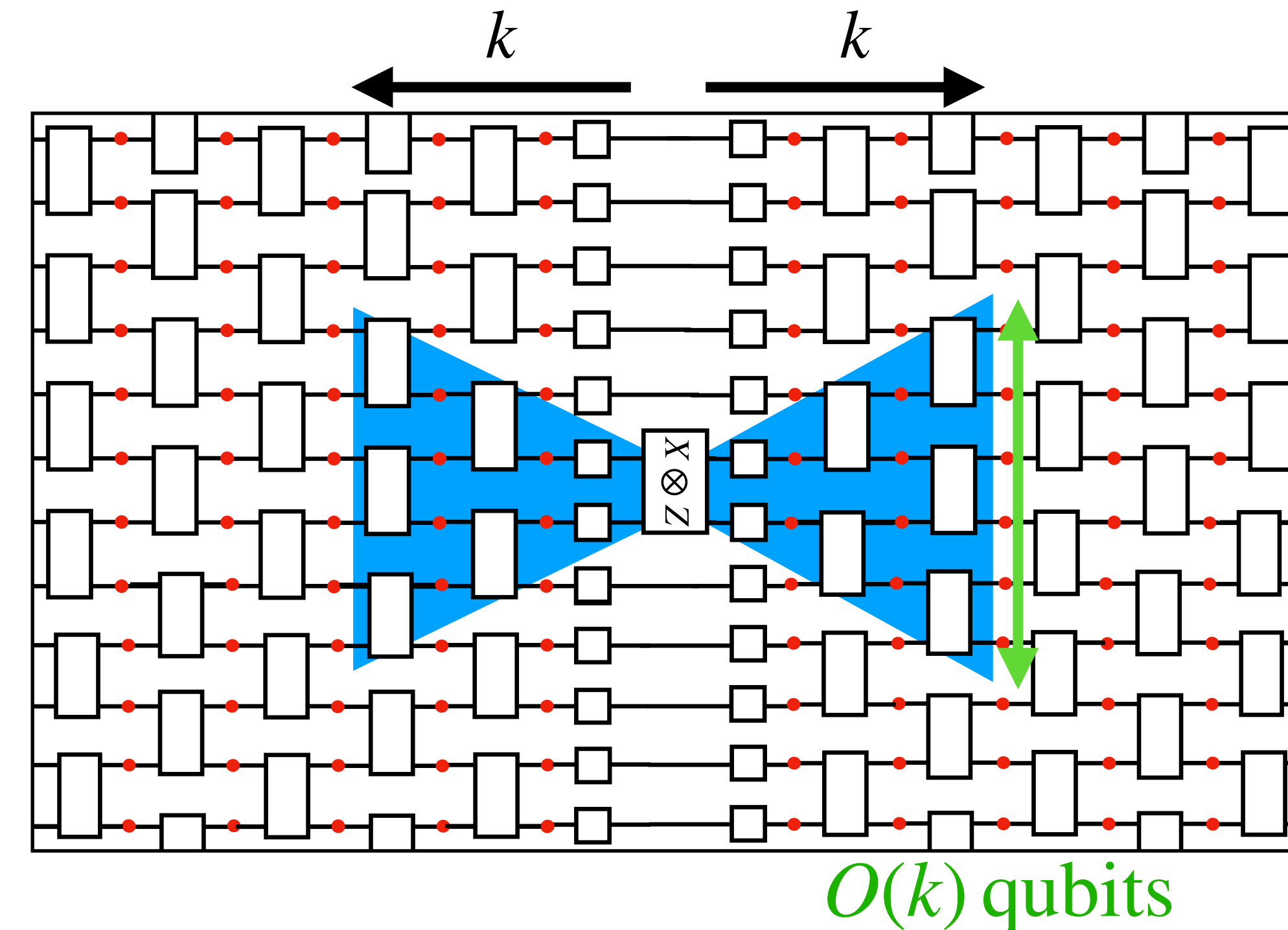
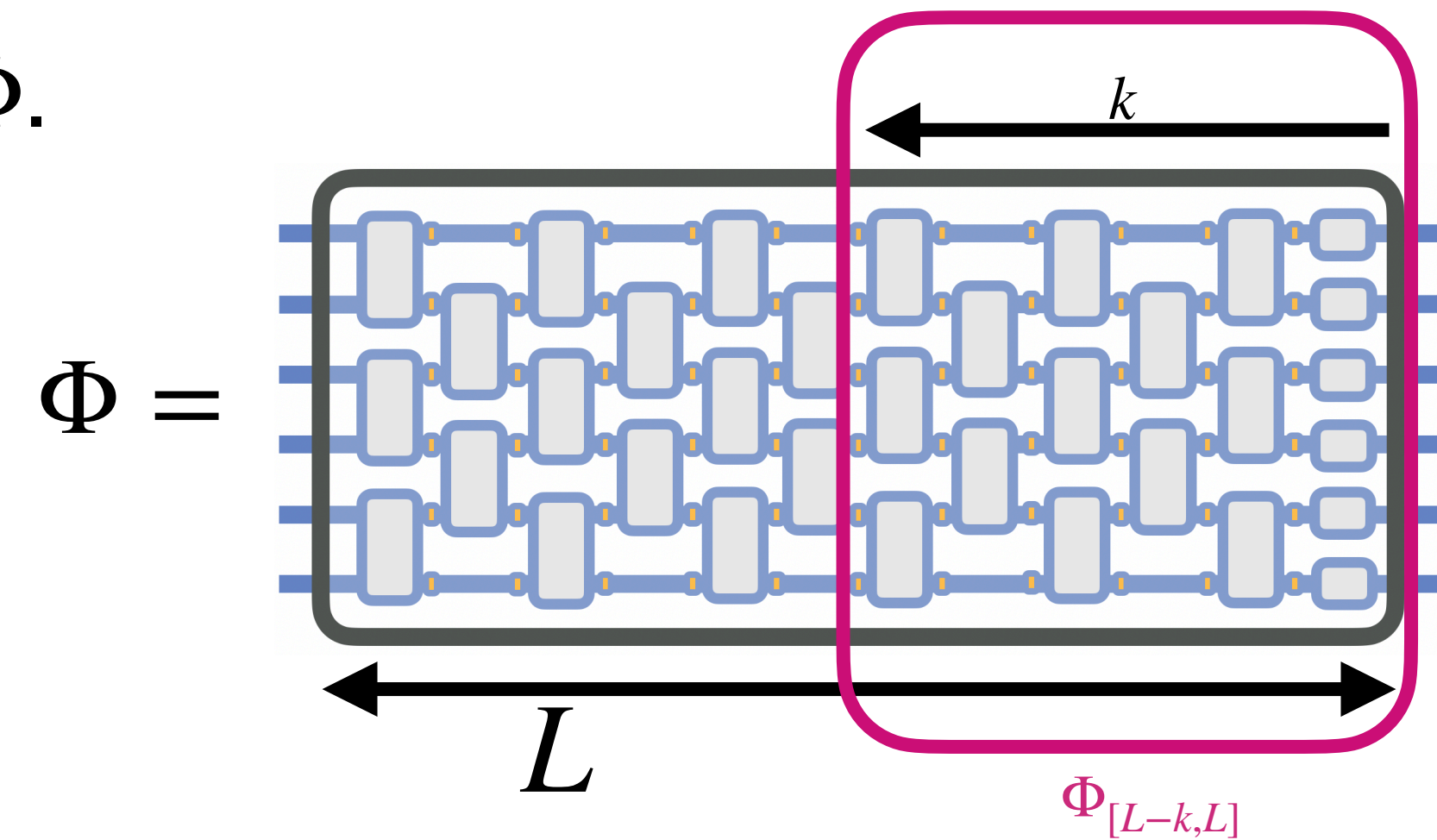
Choosing $k = O(\log(\varepsilon^{-1}))$ suffices to have: $\leq \varepsilon$

• We have $\text{Tr}(P\Phi_{[L-k,L]}(|0^n\rangle\langle 0^n|)) = \text{Tr}(\Phi_{[L-k,L]}^*(P)|0^n\rangle\langle 0^n|)$

If P is local, then this is also local

• Thus, this can be computed efficiently for light-cone arguments.

Computational time (for 1D): $\exp(O(k)) = \text{poly}(\varepsilon^{-1})$

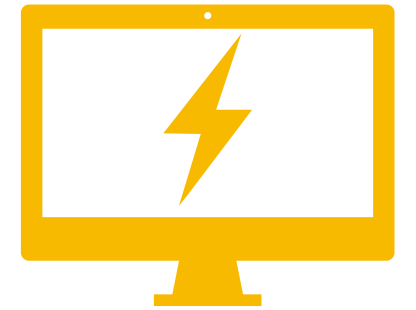


Classical simulation of noisy random quantum circuits expectation values

Theorem.

Given a randomly sampled noisy quantum circuit, for any depth and any noise, there is a classical algorithm to **estimate local Pauli expectation values** up to ε additive precision, with high probability over the choice of the circuit.

The runtime is $\exp(O(\log^D(\varepsilon^{-1})))$, where D is the spatial dimension of the system.

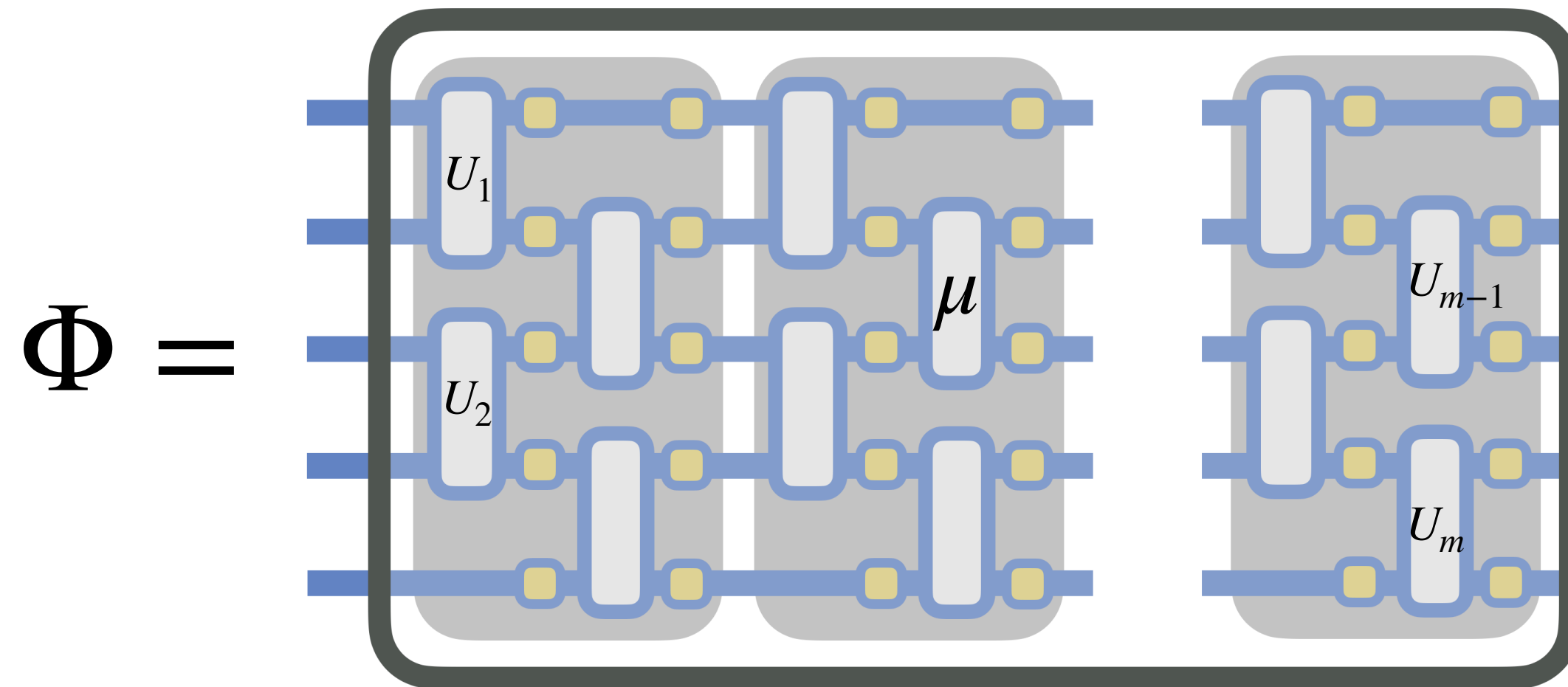


(No dependence by the number of qubits and depth.)

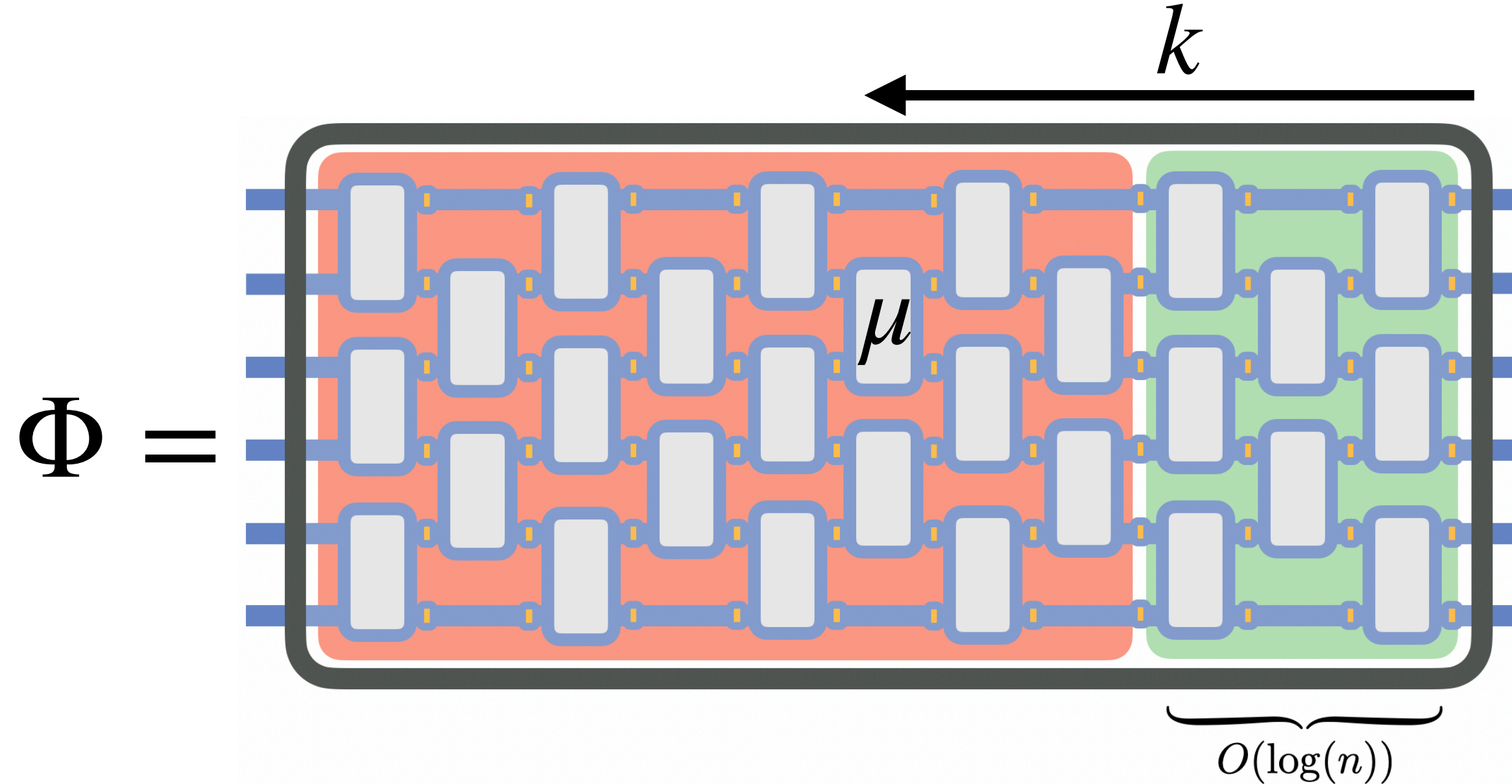
- Thus, for constant ε precision, the algorithm is efficient for any dimensionality.
- For $\varepsilon = \text{poly}(n^{-1})$, the algorithm is efficient for 1D architectures, and quasi-poly for $D > 1$.
- For **global Pauli** expectation values, output zero succeeds with high probability.
- We also give a **condition to verify the success of the classical simulation**:
“Check if the Heisenberg evolved observable is $\sim$ proportional to the Identity”.

Barren plateaus

- **Variational quantum algorithms** encode the problem into $\min_{U_1, \dots, U_m} \text{Cost}(\Phi_{U_1, \dots, U_m})$
- **Barren plateaus (informal)** iff for most choices of $U_1, \dots, U_m$ the cost landscape is ‘flat’ (gradient norm very small).



- Cost functions are made by Pauli expectation values: $C := \text{Tr}(P\Phi(\rho_0))$
- Do variations in the gate μ influence C ? Relevant quantity is $\partial_\mu C$.
If $\text{Var}[\partial_\mu C] \leq \exp(-\Theta(n))$, then the influence is very unlikely.
(a.k.a., the gate is **not trainable**)



Cost function:

$$C := \text{Tr}(P\Phi(\rho_0))$$

☀ **Noiseless [1]:** Barren plateaus for depth $\geq \Omega(n)$. $\text{Var}[C] \leq \exp(-\Theta(n))$,
 (All the gates are not-trainable) $\text{Var}[\partial_\mu C] \leq \exp(-\Theta(n))$.

☀ **Depolarizing noise [2]:** Same of before.

☀ **Non-unital noise [This work]:** No Barren plateaus at any depth for local cost functions.

Only the last $\Theta(\log(n))$ layers are trainable!

$\text{Var}[C] = \Omega(1),$
 $\text{Var}[\partial_\mu C] = \exp(-\Theta(k)).$

[1] Barren plateaus in quantum neural network training landscapes. McClean *et al.* Nature Comm. (2018).

[2] Noise-induced barren plateaus in variational quantum algorithms. Wang *et al.*, Nature Comm. (2021).

Compare with: [3] Beyond unital noise in variational quantum algorithms: noise-induced barren plateaus and fixed points. Singkanipa *et al.*, ArXiv. (2024).

(Lack of) Cost concentration

Theorem.

Let $H := \sum_{P \in \{I, X, Y, Z\}^{\otimes n} \setminus I^{\otimes n}} a_P P$ be an Hamiltonian, ρ_0 an initial state, for any depth of the non-unital noisy circuit Φ , we have:

$$\mathbb{E}_{\Phi}[\text{Tr}(H\Phi(\rho_0))] = 0,$$

$$\text{Var}_{\Phi}[\text{Tr}(H\Phi(\rho_0))] = \sum_{P \in \{I, X, Y, Z\}^{\otimes n} \setminus I^{\otimes n}} a_P^2 \exp(-\Theta(|P|)),$$

where $|P|$ is the Pauli weight.

Corollaries: - Local cost functions do not suffer from exponential concentration.

(For example, $\text{Var}_{\Phi}[\text{Tr}(Z_1\Phi(\rho_0))] = \Omega(1)$.)

- Global cost functions suffer from exponential concentration.

(For example, $\text{Var}_{\Phi}[\text{Tr}(Z^{\otimes n}\Phi(\rho_0))] = \exp(-\Theta(n))$.)

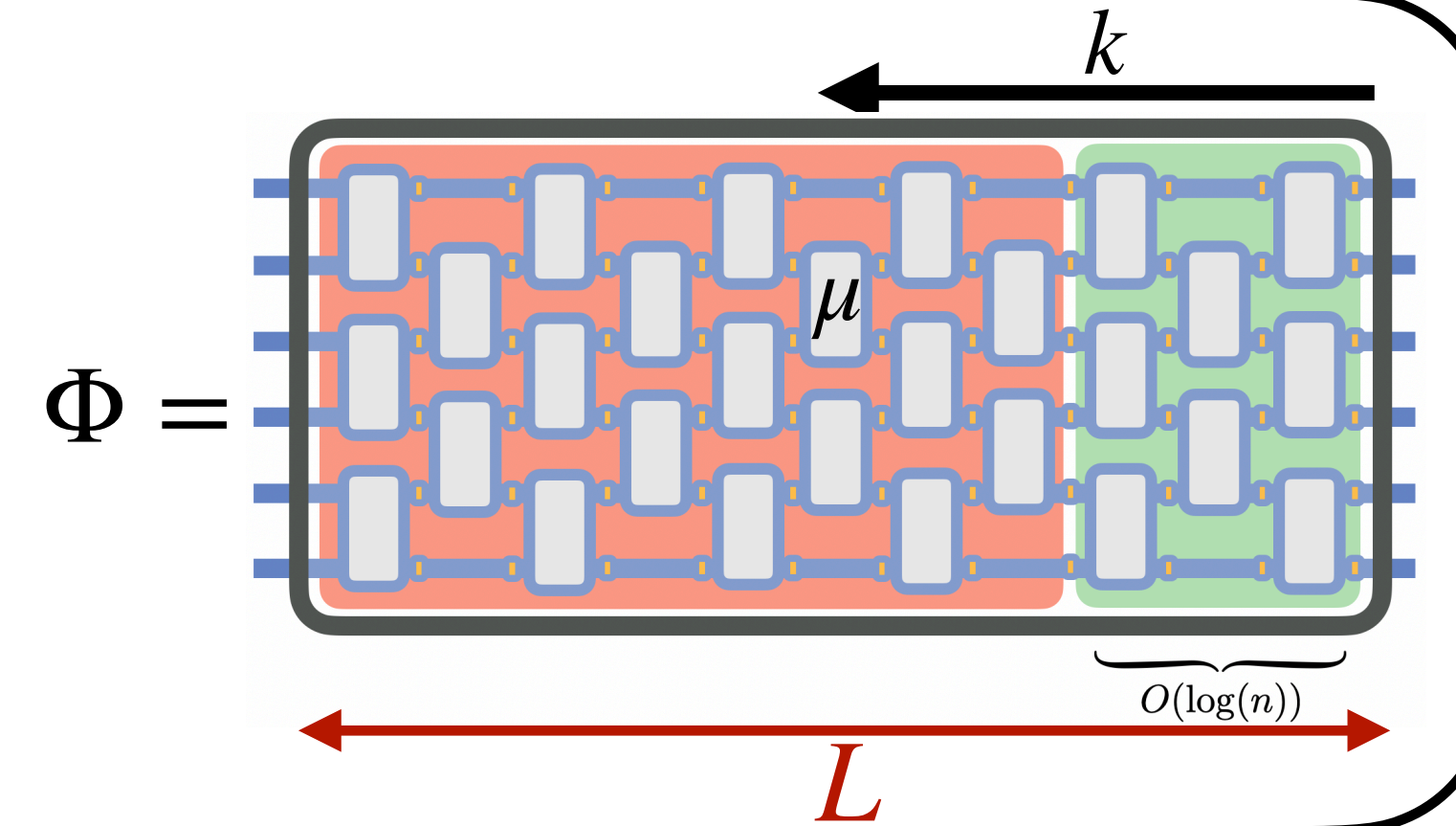
Partial derivatives

Theorem.

Let $C = \text{Tr}(\Phi(\rho_0)P)$, where P is a Pauli, at any depth and noise:

$$\mathbb{E}_{\Phi}[\partial_{\mu} C] = 0$$

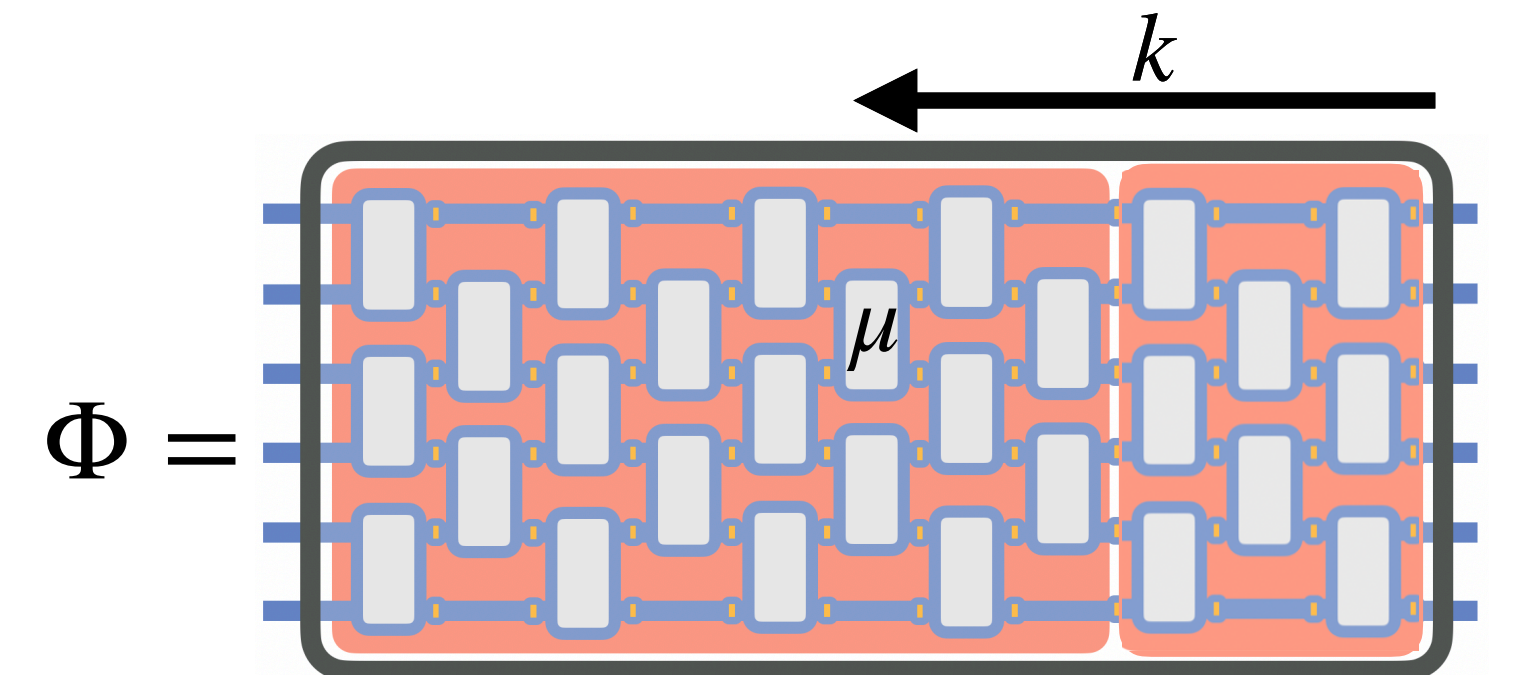
$$\text{Var}_{\Phi}[\partial_{\mu} C] \leq \exp(-\Theta(k + |P|)),$$



- If P is local, then for any non-unital noise $\text{Var}_{\Phi}[\partial_{\mu} C] \geq \exp(-\Theta(k))$.

Corollary: $\mathbb{E}_{\Phi}[\|\nabla C\|_2^2] = \Omega(1)$.

- If P is global, then for any gate $\text{Var}_{\Phi}[\partial_{\mu} C] = \exp(-\Theta(n))$.



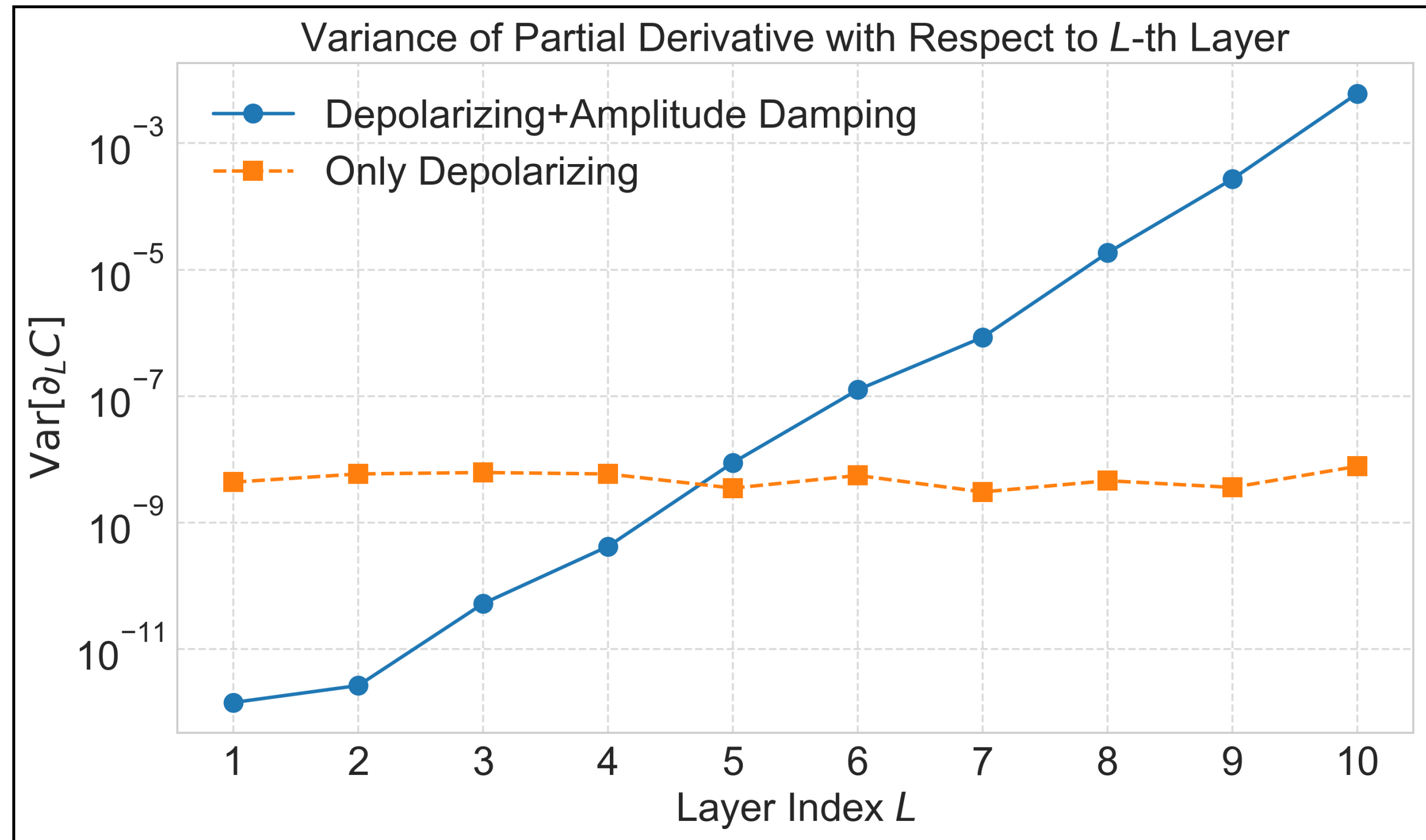
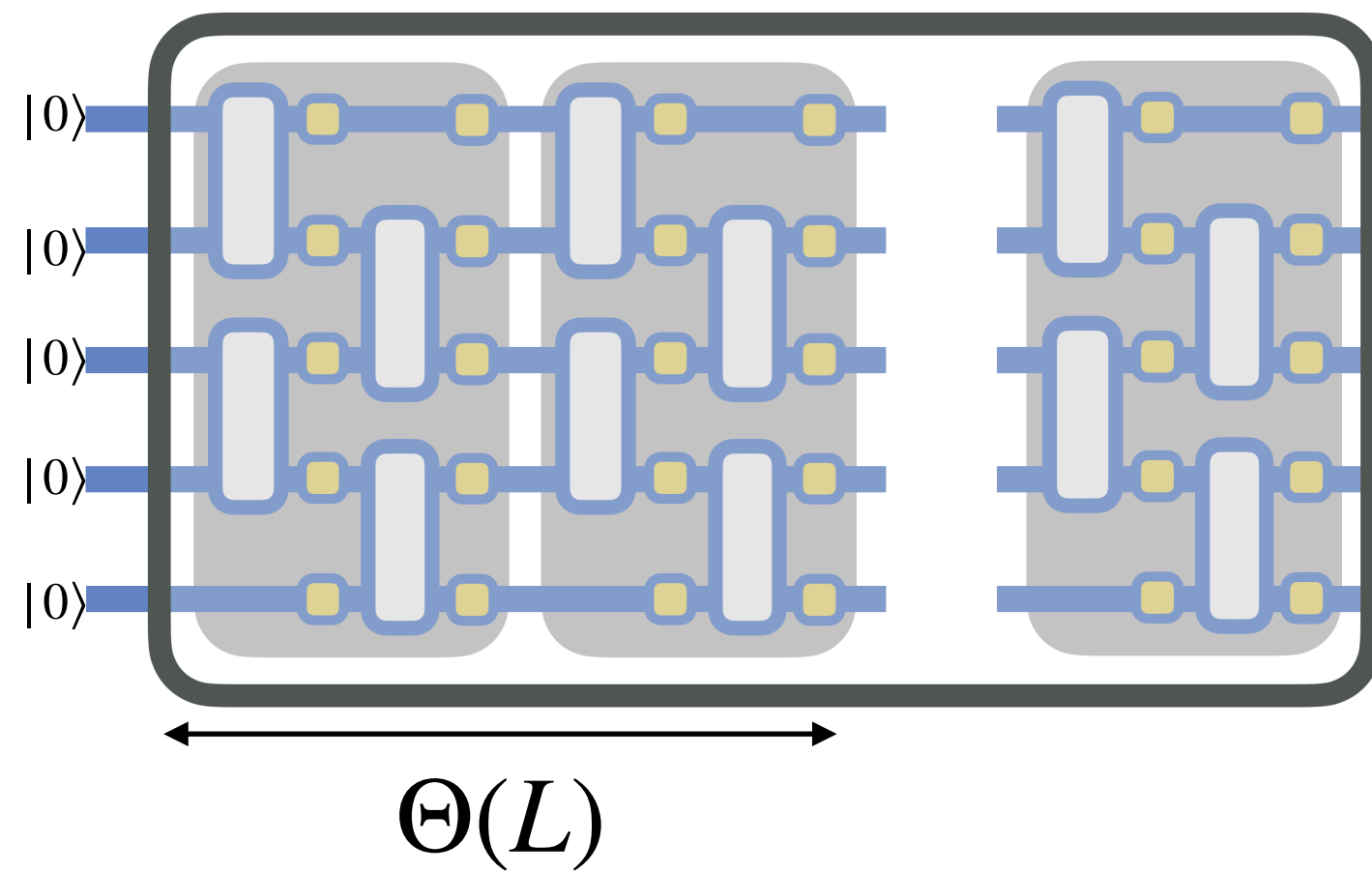
- For the particular case of unital noise, our tools imply: $\text{Var}_{\Phi}[\partial_{\mu} C] \leq \exp(-\Theta(L + |P|))$, which improves upon [1].

Proof ingredients



- Work in the Heisenberg picture. $\text{Tr}(P\Phi(\rho_0)) = \text{Tr}(\Phi^*(P)\rho_0)$
- 2-design properties (of the 2-qubits gates).
- ‘Normal form’ of noisy channels. Any 1-qubit channel is unitarily equivalent to a channel parametrized by:
$$(D_X, D_Y, D_Z), (t_X, t_Y, t_Z) \in \mathbb{R}^3$$
$$\mathcal{N}(I) = I + t_X X + t_Y Y + t_Z Z$$
$$\mathcal{N}(X) = D_X X$$
$$\mathcal{N}(Y) = D_Y Y$$
$$\mathcal{N}(Z) = D_Z Z$$

Numerical experiments confirm our results for more structured circuits.



Conclusions

- Any amount of **non-unital noise induces lack of Barren plateaus**, but it ‘truncates’ most circuits to **effectively log-depth**, allowing **efficient classical simulation** in estimating Pauli expectation values.
- **Unless** we carefully **engineer the circuits to take advantage of the noise**, it is **unlikely** that **non-unital** noisy circuits are **preferable over depolarizing** noisy circuits (which are restricted to log depth).
- Assuming overly simplistic noise models could be misleading.

Open questions

- Simulating Pauli expectation values of noisy random quantum circuits with $\text{poly}(n^{-1})$ accuracy in 2D and all-to-all connectivity in polynomial time.
- Complexity of classical simulation of random quantum circuit sampling with non-unital noise [1].

(The depolarizing case was addressed in [2])

Thanks a lot for your attention!

[1] Effect of non-unital noise on random circuit sampling, Fefferman *et al*, QIP (2023).

[2] A polynomial-time classical algorithm for noisy random circuit sampling, Aharonov *et al.*, STOC (2023)